# Assessing the group identification

What chance means, how real pairs differ from it,
and how the result moves with the parameters

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Companion of the note *Group flights in free flight* (same folder)

**The method.** Two pilots are in contact when one of them passes within $\varepsilon_h = 500$ m and $\varepsilon_z = 300$ m of where the other was at most $\delta = 2$ min before or after; headings are compared only when both glide ($\theta_{\max} = 45°$) and gaps shorter than $g = 120$ s are joined. A contact counts when it lasts at least $\tau^* = 38.1$ min. With these values 21.7% of the paraglider flights and 6.9% of the hang-glider flights fly in a group at some point, in 17 169 episodes.

**Chance.** A pair of the same day is made independent by shifting one flight in time, by 58 min in the median (Section 2). Its contacts are what route, terrain and day give without synchrony; $\tau^*$ is the duration at which they are one in ten ($q = 0.1$) of the contacts observed. The length of the shift matters: with only shifts under 30 min $\tau^*$ would be 53 min, with only shifts over an hour 23 min.

**Quality.** Real pairs differ from chance in every view: in duration (most short contacts are chance, few long ones), in delay (5.6 times chance at a few seconds) and in distance (10 times chance within 100 m at the same instant). The contacts of at least 90 min have a median delay of 25 s (Section 3).

**Parameters.** $\delta$ is measured on the delay profile and $\tau^*$ follows from $q$. The profiles of $\varepsilon_h$, $\varepsilon_z$ and $g$ have no edge, so these three are definitions; changing them moves the share of time in a group by up to a factor three, mostly through $\tau^*$ (Section 4).

**Where.** `/Volumes/SSD_DISANTE/xc-lab/group-identification/`: the chosen run in `runs/d120_eh500_ez300_th45_g120_q0.1`, the profiles in `delays/`, `separation/` and `tolerances/`, each with a `PARAMETERS.md`.

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# 1 The parameters

The method of the companion note ran on all the cleaned flights of the FFVL archive, 162 485 flights of 6 209 dates; the profiles use a sample of 300 dates. It has three kinds of parameters (Table 1). The *tolerances* say when two positions match: $\delta$, $\varepsilon_h$, $\varepsilon_z$, $\theta_{\max}$, $g$. The *duration* $\tau^*$ says how long a match must last to be a contact; it is computed from the share $q$ of chance contacts we accept. The other *settings* make the computation exact or fast and are fixed by physical bounds.

| Parameter | Value | How it was set |
|---|---|---|
| $\delta$, largest delay | 2 min | measured on the delay profile, Section 3.2 |
| $q$, accepted false discovery rate | 0.1 | trade-off, Section 5.2 |
| $\tau^*$, shortest contact | 38.1 min | computed from $q$ and the null model, Section 2.2 |
| $\varepsilon_h$, $\varepsilon_z$ | 500 m, 300 m | definitions: profiles without an edge, Section 4 |
| $g$, longest gap joined | 120 s | definition: gaps of real and chance contacts alike, Section 4 |
| $\theta_{\max}$ | 45° | no effect on the result, Section 4.2 |
| $\Delta$, $h_{\max}$, $R_{\text{dep}}$, $v_{\max}$, $R_s$, $r_{\min}$ | see Section 6 | physical bounds and exactness |

**Table 1:** The parameters and how each was set.

# 2 What chance means

## 2.1 The null model

Two pilots who never interact still meet: they take off from the same site, follow the same ridges and valleys and climb in the same house thermals. The null model of the companion note builds such pairs from the data. On each date, up to 200 pairs of flights whose takeoffs are less than $R_s = 10$ km apart are drawn; in each, one flight $j$ is replaced by its copy shifted in time, $\mathbf{x}_j^s(t) = \mathbf{x}_j(t + r)$. The shift is $r = (k_j^0 - k_i^0) - \vartheta$, with $\vartheta$ drawn from the takeoff delays of the pairs of that date, so the copy of $j$ takes off $\vartheta$ after $i$; only $|r| \geq r_{\min} = 2\delta + \Delta = 245$ s is accepted. A pair enters once, with the roles of $i$ and $j$ drawn at random.

What the shifted pair keeps: the route, the terrain, the date, and the takeoff delays of the day as a distribution. What it destroys: every correspondence within $\delta$, since $|r| > 2\delta$. The shift is far larger than its minimum (Figure 1): 58 min in the median, and 90% of the shifts lie between 8 min and 4.4 h. So the copy also flies in the weather of another hour, with other thermals working and another cloud base.

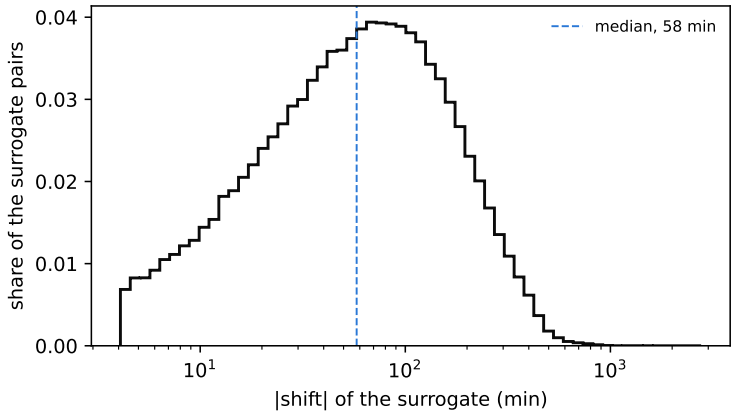



**Figure 1:** The time shifts $|r|$ of the 248 741 surrogate pairs of the chosen run, whole archive.

Chance is therefore an operational notion: the contacts that the same routes give without synchrony. Two consequences follow.

- Pilots who use the same thermals at the same time without following each other are in synchrony with the weather of the hour, which the shift removes: their contacts count as real.
- A pilot who follows another deliberately, but more than $\delta$ behind, matches the other only beyond $\delta$: the real pair never shows this contact, while a surrogate whose shift brings that delay within $\delta$ counts it as chance.

### 2.2 From the surrogates to the false discovery rate

On each date $d$:

1. $\mathcal{P}_d$ is the set of the pairs $(a, b)$, $a < b$, of flights of the date with takeoffs less than $R_s$ apart.
2. The exposure of a pair is the time both fly after departure, $h_{ab} = \max\big(\min(k_a^{\text{fin}}, k_b^{\text{fin}}) - \max(k_a^{\text{dep}}, k_b^{\text{dep}}),\, 0\big)$, and $H_d = \sum_{\mathcal{P}_d} h_{ab}$.
3. $\mathcal{S}_d$ is the set of the surrogate pairs $(i, j, r)$; $S_d$ is their exposure, the same sum with $j$ shifted: $S_d = \sum_{\mathcal{S}_d} \max\big(\min(k_i^{\text{fin}}, k_j^{\text{fin}} - r) - \max(k_i^{\text{dep}}, k_j^{\text{dep}} - r),\, 0\big)$.
4. $M_d(\tau)$ is the number of contact intervals of the surrogate pairs lasting at least $\tau$, and $\rho_{0,d}(\tau) = M_d(\tau)/S_d$ the chance intervals per second of exposure.

Over the archive, with $N(\tau)$ the number of real intervals of the pairs of every $\mathcal{P}_d$ lasting at least $\tau$:

$$C(\tau) = \sum_d \rho_{0,d}(\tau)\, H_d, \qquad \widehat{\text{FDR}}(\tau) = \frac{C(\tau)}{N(\tau)}, \qquad \tau^* = \min\big\{\tau :\ \widehat{\text{FDR}}(\tau) \le q\big\}. \tag{1}$$

$C(\tau)$ is the number of intervals lasting at least $\tau$ that chance alone would give the real pairs. It is summed date by date because the surrogates are a sample of the pairs of each date; dates without surrogates add to $N$ only. $\tau$ runs over the grid $\{0, 5, \ldots, 14\,400\}$ s.

| Surrogates | pairs | $\widehat{\mathrm{FDR}}$(10 min) | $\widehat{\mathrm{FDR}}$(30 min) | $\tau^*$ (min) | 5–95% over dates |
|---|---|---|---|---|---|
| all shifts | 241 824 | 0.48 | 0.16 | 38.2 | 37.0–41.5 |
| $\lvert r\rvert < 30$ min | 69 444 | 0.73 | 0.31 | 53.3 | 49.7–72.0 |
| $\lvert r\rvert \geq 60$ min | 119 687 | 0.26 | 0.05 | 23.0 | 21.1–24.4 |

**Table 2:** The false discovery rate and $\tau^*$ ($q = 0.1$, $\delta = 2$ min) with the surrogates of one class of shifts, on the 2 737 dates with surrogates in each class.

### 2.3 How much the shift matters

The false discovery rate was recomputed with only the surrogate pairs of each date whose shift falls in one class, each class with its own exposure $S_d$ and intervals. The classes are compared on the 2 737 dates (of 6 209) that have surrogates in each of them (Table 2).

Short shifts give 2.8 times the chance of long ones at 10 min and 5.7 times at 30 min, and $\tau^*$ goes from 23 min with shifts of an hour or more to 53 min with shifts under 30 min; with all the shifts it is 38 min, as on the whole archive. Short shifts keep the weather of the hour, but they also bring within $\delta$ the pilots who follow each other a few minutes apart, and the delay profile shows that these are many: real correspondences are still 2 to 3 times chance between 195 and 500 s (Section 3.2). The two effects cannot be separated here. The length of the shift decides what counts as chance and moves $\tau^*$ by more than a factor two; we keep the shifts that follow the takeoff delays of the day.

## 3 Real pairs against chance

Three views compare the real pairs with their surrogates: how long their contacts last, at which delay the two pilots match, and how far apart they are at the same instant.

### 3.1 Duration

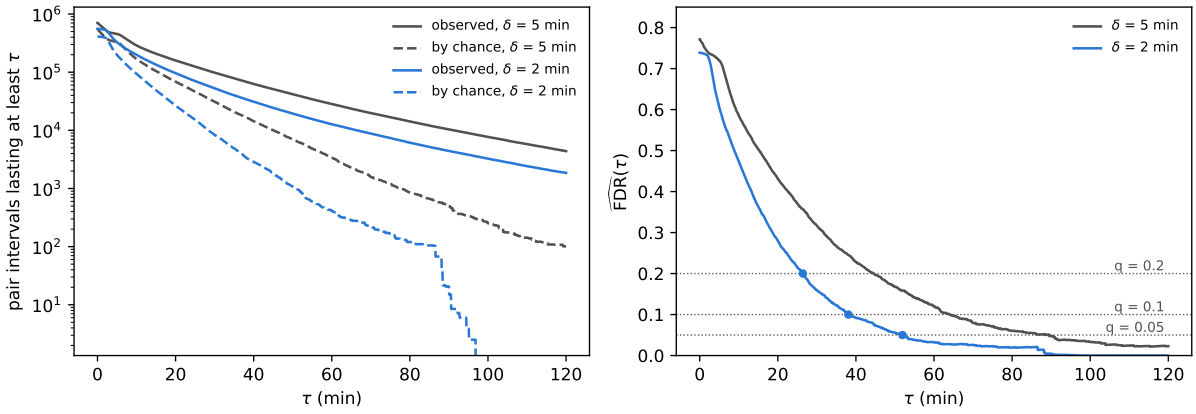



**Figure 2:** Left: real pair intervals lasting at least $\tau$ (solid) and those chance alone gives (dashed), with $\delta = 5$ min (grey) and $\delta = 2$ min (blue), on the whole archive. Right: their ratio, the false discovery rate of Eq. (1); dots mark $\tau^*$ for $q = 0.05$, 0.1 and 0.2 with $\delta = 2$ min.

Most contacts are chance (Figure 2): 74% of all pair intervals with $\delta = 2$ min (77% with 5 min), 47% of those lasting 10 min, 16% of those lasting 30 min. Short contacts therefore say little; only long ones are mostly real. This is why $\tau^*$ is tens of minutes: it is the duration at

which the real contacts outnumber chance by the factor $1/q$. Resampling the dates 200 times moves $\tau^*$ by a few minutes (Table 5), so the curve, and not the noise of the surrogates, sets it.

### 3.2 Delay

For every delay $l$, the correspondences $(t, t+l)$ of the real pairs and of the surrogate pairs were counted, per second of exposure, with a window of 15 min on 300 random dates (58 910 h of real exposure):

$$R(l) = \frac{\text{real correspondences at delay } l \text{ per second of exposure}}{\text{chance correspondences at delay } l \text{ per second of exposure}}. \tag{2}$$

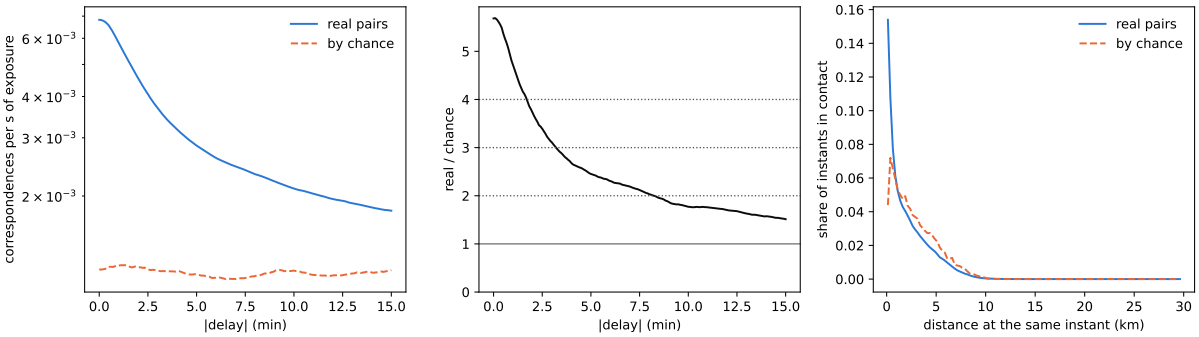



**Figure 3:** The delay profile on 300 dates with a 15 min window. Left: correspondences per second of exposure at each delay, real and by chance. Middle: their ratio $R(l)$ of Eq. (2), with the levels 4, 3 and 2. Right: the distance between the two pilots at the same instant when one is on the path of the other.

Chance is flat in the delay, as it should: a shifted copy has no preferred delay. The real pairs peak at a few seconds, where they have 5.6 times the correspondences of chance, and decay: $R$ falls below 4 at 105 s, below 3 at 195 s and below 2 at 500 s, and is still 1.5 at 15 min (Figure 3). The excess has two parts. The fast one, within 2–3 min, is pilots flying together. The slow one, which has not ended at 15 min, is pilots who use the same thermals at about the same time without following each other, a synchrony that the shifts of the surrogates remove. A rule such as "the delay that holds 95% of the excess" would pick 13 min, because it counts the slow tail; that is why we read the ratio instead.

### 3.3 The delay inside the contacts

The delay profile counts correspondences; this view looks at the contacts themselves. Every pair interval has a delay. At each instant $t$ of $i$ in the interval with $m_{ij}(t) = 1$ and neither pilot in search, $l^*(t)$ is the delay of the closest correspondence that passes (smallest horizontal distance); the delay of the interval is the median of $l^*(t)$, positive when $j$ follows $i$. Its absolute value was computed for every pair interval of the run with $\delta = 5$ min, whose window is wide enough to show where the delays fall: 965 770 intervals, of which 169 517 have all their instants in search and no delay.

The longer the contact, the shorter the delay (Table 3, Figure 4). A pair that only shares route and weather matches at any delay of the window, and its closest correspondence is often at the largest ones: short intervals pile up at the edge, 29% of those under 10 min. The contacts of at least 90 min, all above $\tau^*$ of this run (89.5 min), have a median delay of 25 s; 90% of them lie within 2 min and only 6 of 10 734 at the edge. Long contacts are pilots a few tens of seconds apart on the same path, which is what $\delta = 2$ min keeps (Section 5.1).

| Duration of the interval | intervals | median \|delay\| | within 2 min | at the edge |
|---|---|---|---|---|
| under 10 min | 436 575 | 210 s | 32% | 29% |
| 10–30 min | 246 587 | 160 s | 39% | 12% |
| 30–90 min | 102 357 | 80 s | 64% | 2.1% |
| 90 min or more | 10 734 | 25 s | 90% | 6 intervals |

**Table 3:** The delay of the pair intervals by their duration ($\delta = 5$ min, whole archive). At the edge: $|\text{delay}| \geq 285$ s, the last 15 s of the window.

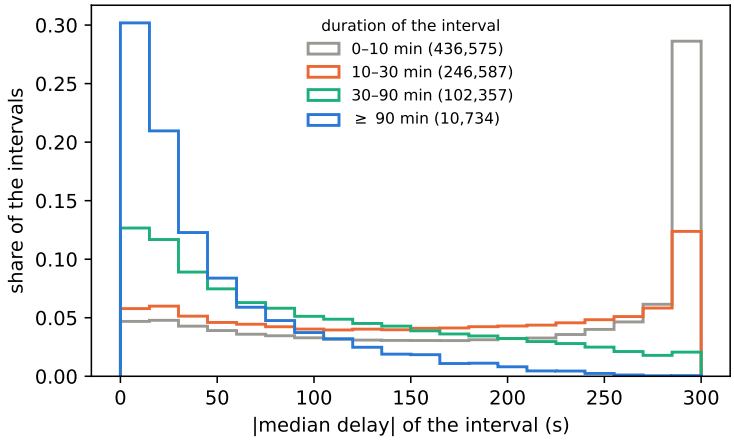



**Figure 4:** Distribution of the |delay| of the pair intervals, by their duration ($\delta = 5$ min, whole archive).

### 3.4 Distance at the same instant

In analogy with the delay profile, the distance between the two pilots at the same instant was counted at every grid instant when both fly after departure, for all the real pairs and for the surrogates, on the dates of Figure 3 that have surrogates (210 of 300, 58 901 h of real exposure). Unlike the right panel of Figure 3, no correspondence is required: every instant of every pair counts.

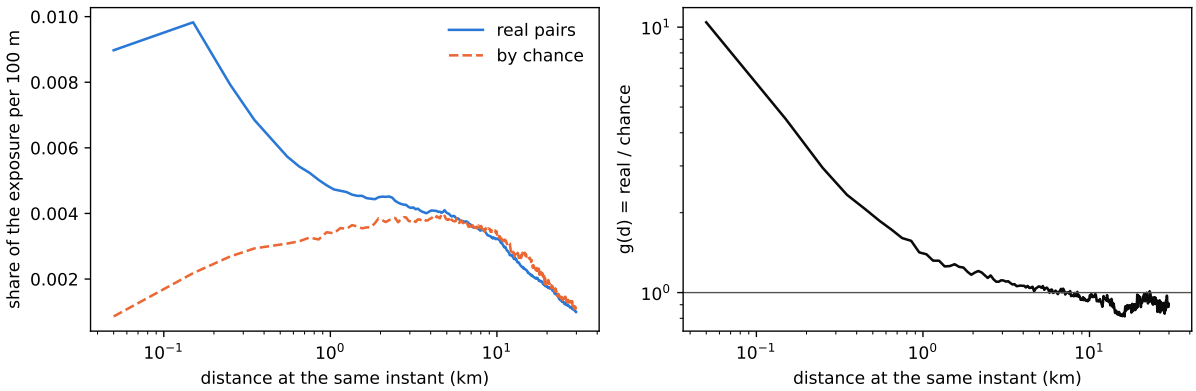



**Figure 5:** Distance between the two pilots at the same instant, over all the time both fly, on 210 dates. Left: share of the exposure in each 100 m bin, real pairs and chance. Right: their ratio $g(d)$.

Real pairs are much closer than chance at short distance and have no edge (Figure 5): $g(d) = 10.4$ within 100 m, below 2 from 500 m, below 1.5 from 900 m, below 1.1 from 2.7 km and below 1 from 5.7 km. Real pairs spend 6.6% of their time together within 1 km of each other, against 2.8% by chance. Beyond 6 km $g$ falls slightly below 1, because both distributions

sum to one.

The excess reaches several kilometres, beyond the visual range of 1.5 km taken in the companion note. Since the surrogates also change the hour, this excess may hold the weather of the hour as well as sight, so it bounds neither the visual range nor the range of following.

# 4 The tolerances

## 4.1 Profiles against chance

In analogy with the delay profile, with $\delta = 2$ min and the null model of the chosen run, at every instant $t$ of pilot $i$ on the 210 dates of Figure 5:

- the smallest horizontal distance between $i$ at $t$ and $j$ at $t + l$, $|l| \leq \delta$, over the delays with $|\Delta z| < \varepsilon_z$ and headings that pass (left of Figure 6);
- the smallest height difference over the delays with horizontal distance $< \varepsilon_h$ and headings that pass (middle);
- the gaps between consecutive instants of contact of a pair (right).

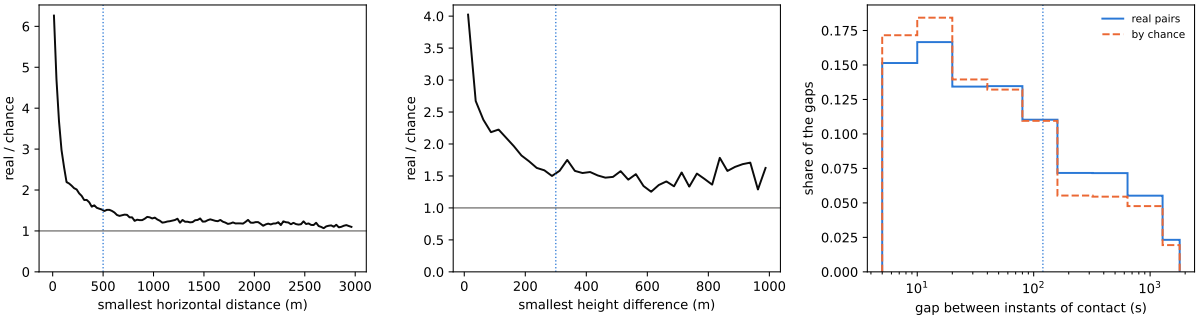



**Figure 6:** The tolerance profiles on 210 dates. Left and middle: real pairs over chance, per second of exposure, by the smallest horizontal distance and the smallest height difference to the path of the other pilot within $\delta$; dotted, the values in use. Right: the lengths of the gaps between instants of contact, as a share of all gaps (the 8% of real and 9% of chance gaps longer than 30 min are not drawn); dotted, $g$.

None of the three has an edge (Figure 6). The horizontal ratio is 6.3 within 25 m, below 2 from 250 m, below 1.5 from 500 m, and still 1.13 between 2.5 and 3 km. The vertical ratio is 4.0 within 25 m and below 2 from 150 m; from 275 m on it stays near 1.5 (1.51 on average up to 1 km), the excess that the horizontal condition alone gives. Both values in use sit where real pairs are about 1.5 times chance.

Real pairs have 1.46 times the gaps of chance per hour, because they have more contact, but the lengths of the gaps are the same: the shares of each class differ by at most 2 points. The gaps come from how pilots fly in and out of each other's path, the same in real and in chance contacts, so the data do not single out $g$.

## 4.2 One parameter at a time

The companion note asks for a plateau: values of the tolerances around which the result does not change. We changed one parameter at a time on 300 random dates, recalibrating $\tau^*$ each time, with $\delta = 5$ min (Table 4). $\theta_{\max}$ changes nothing: contacts in glide are few. The others move the fraction of flight time in a group, $f_G$, by up to a factor three, mostly through $\tau^*$:

there is no plateau, in agreement with the profiles. On 300 dates $\tau^*$ is noisy (126 min with the starting values against 89.5 on the whole archive, and six settings stop at the same surrogate interval); a conclusive check needs the whole archive for each setting, about one hour each. It was not repeated with $\delta = 2$ min.

| Setting ($\delta = 5$ min unless stated) | $\tau^*$ (min) | $f_G$ | episodes | flights in a group |
|---|---|---|---|---|
| starting values | 126.1 | 0.034 | 98 | 210 |
| $\varepsilon_h = 250$ m | 75.1 | 0.058 | 253 | 537 |
| $\varepsilon_h = 1000$ m | 126.8 | 0.065 | 162 | 372 |
| $\varepsilon_z = 150$ m | 126.1 | 0.020 | 61 | 125 |
| $\varepsilon_z = 600$ m | 126.1 | 0.044 | 122 | 268 |
| $\delta = 2$ min | 42.6 | 0.106 | 600 | 1 170 |
| $\delta = 10$ min | 128.4 | 0.062 | 153 | 365 |
| $\theta_{\max} = 30°$ | 126.1 | 0.034 | 97 | 208 |
| $\theta_{\max} = 90°$ | 126.1 | 0.034 | 98 | 210 |
| $g = 60$ s | 75.8 | 0.087 | 332 | 732 |
| $g = 300$ s | 126.1 | 0.055 | 141 | 318 |

**Table 4:** One parameter at a time on 300 random dates (7 051 flights), $q = 0.05$.

# 5 The parameters set by the data

## 5.1 The delay $\delta$

We take $\delta = 2$ min: within it the real correspondences are at least 3.7 times chance (Section 3.2), and it contains nine in ten of the long contacts (Section 3.3). Shortening the delay also shortens $\tau^*$, since fewer chance matches enter: from 89.5 to 51.9 min at $q = 0.05$ (Table 5).

The delay also bounds how far apart two pilots in contact can be at the same instant: at most $\varepsilon_h + v_{\max}\delta \approx 0.5 + 27.8 \times 120/1000 \approx 3.8$ km. A separation of 6 km, the largest we want to call a group, is never reached by one pair; a group, a chain of pairs, can be wider. Even with the 15 min window, 58% of the instants in contact have the two pilots within 2 km and 94% within 6 km (Figure 3, right).

## 5.2 The false discovery rate $q$

$q$ is the share of chance contacts we accept among those we keep. A smaller $q$ keeps only very long contacts and few groups, all of them real; a larger $q$ sees more groups, with more chance among them. Table 5 and Figure 7 give the trade-off.

| $\delta$ | $q$ | $\tau^*$ (min) | 5–95% over dates | para in a group | hang in a group | episodes |
|---|---|---|---|---|---|---|
| 5 min | 0.05 | 89.5 | 79.5–91.5 | 8.9% | 2.1% | 6 090 |
| 2 min | 0.05 | 51.9 | 49.1–53.3 | 13.7% | 3.8% | 10 553 |
| 2 min | 0.1 | 38.1 | 37.0–41.1 | 21.7% | 6.9% | 17 169 |
| 2 min | 0.2 | 26.4 | 25.2–27.2 | 31.7% | 13.9% | 26 504 |

**Table 5:** $\tau^*$ and the groups it gives. Flights with a logger of at most 10 s and a position after departure: 154 598 paragliders and 6 039 hang gliders.

We take $q = 0.1$: at most one contact in ten is chance, and $\tau^* = 38.1$ min. Going to 0.2 would lower $\tau^*$ to 26 min but would let chance make one group in five.

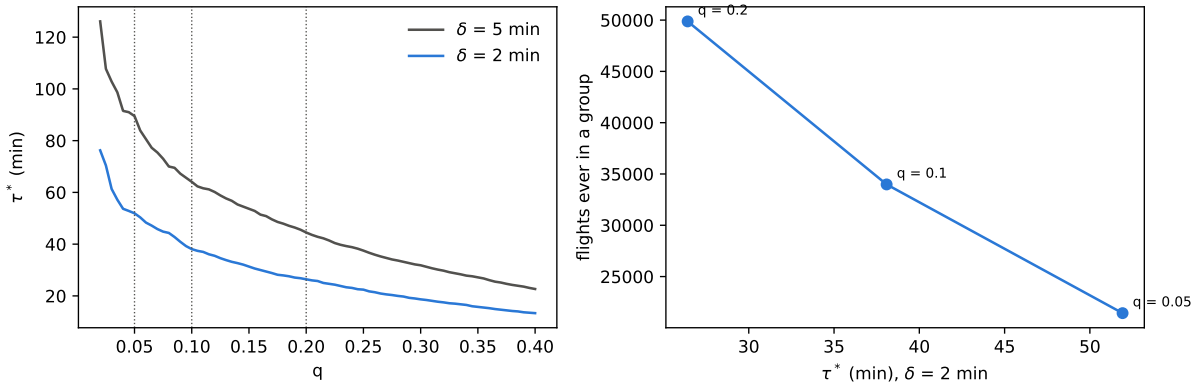



**Figure 7:** Left: $\tau^*$ as a function of $q$, for the two delays. Right: flights ever in a group against $\tau^*$, for $q = 0.05$, 0.1 and 0.2 with $\delta = 2$ min.

## 6 Fixed settings

- $\Delta = 5$ s, the common grid: a pilot is at most 69 m from a grid position at 100 km/h, against $\varepsilon_h = 500$ m.
- $h_{\max} = 10$ s: positions are interpolated only between fixes at most 10 s apart; the flights of coarser loggers are left out of the statistics.
- $R_{\text{dep}} = 3$ km: contacts around the takeoff, before the first exit from this disc, are not counted.
- $v_{\max} = 100$ km/h, the speed of the candidate filter: pieces of track end at every faster step, so the filter never misses a correspondence (checked against every pair of instants on synthetic days). The profiles of Section 4 widen the filter to 3 km, the largest distance they count.
- $R_s = 10$ km, $r_{\min} = 2\delta + \Delta$, at most 200 surrogate pairs per date: the null model of Section 2.

## 7 Limits

- Chance is defined by the null model of Section 2. Its shifts, about an hour, also remove the synchrony of pilots with the weather of the hour, so what it calls real includes pilots who share the same thermal at the same time without following each other: the "same air" notion of the companion note, which is not proof of interaction. $\tau^*$ is 23 min with shifts over an hour and 53 min with shifts under 30 min (Section 2.3).
- The profiles of Sections 3.2, 3.4 and 4 are on one sample of 300 dates.
- The plateau of Section 4.2 is on 300 dates and with the old delay.
- The data are the cleaned tracks of the old repository (soaring, tag `legacy-final`); the phases needed by the heading condition exist on 1 Hz flights only.